Campbell B

Weekly Presentation

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Campbell B

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1. Strategies
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2026/Spring

Basic Strategy

- Total of three prompts were used to explore basic strategy.
- Prompt 1 - Basic strategies for a team project of no finance background.
- Prompt 2 - Basic strategies for a team project.
- Prompt 3 - Basic strategies for a team project which involves NLP.
- All the three prompts are slight variations of each other.
- All the three prompts were passed to ChatGPT 5.2 and Gemini 3.

Prompt 1

- I am involved in a group project to develop trading strategies using AI, implement and backtest them in python. As I have no finance background, suggest trading strategies that I could start with. Also guide on implementing the same in python and how to get relevant data to backtest the strategies. The aim of the project is to start with a particular strategy (could be basic) and then iterate to refine the strategy and add more components. The budget of the project is 0, so suggest data which can be used in python for backtesting without any cost.

Prompt 2

- I am involved in a group project to develop trading strategies using AI, implement and backtest them in python. Suggest trading strategies that I could start with. Also guide on implementing the same in python and how to get relevant data to backtest the strategies. The aim of the project is to start with a particular strategy (could be basic) and then iterate to refine the strategy and add more components. The budget of the project is 0, so suggest data which can be used in python for backtesting without any cost.

Prompt 3

- I am involved in a group project to develop trading strategies using AI, implement and backtest them in python. The finance background of the group is little. Suggest trading strategies that I could start with which involves sentiment analysis or language processing. Also guide on implementing the same in python and how to get relevant data to backtest the strategies. The aim of the project is to start with a basic version of a strategy and then iterate to refine the strategy and add more components. The budget of the project is 0, so suggest data which can be used in python for backtesting without any cost.

Prompt 1 Results - ChatGPT 5.2

- Strategies that are AI-friendly to iterate on
  - Time-series momentum (trend-following) — simplest and most extensible
  - Cross-sectional momentum (relative strength)
  - Mean reversion (RSI / Bollinger / z-score)
  - Simple statistical arbitrage on pairs (pairs trading)

Prompt 1 Results - Gemini 3

- Strater strategies
  - Mean Reversion (RSI-Based)
  - Pairs Trading (Statistical Arbitrage)
  - Breakout Strategy (Donchian Channels)

Prompt 2 Results - ChatGPT 5.2

- Starter strategies that are simple to implement
  - Time-Series Momentum (Trend Following)
  - Cross-Sectional Momentum (Relative Strength)
  - Mean Reversion (Pairs Trading or Z-Score Reversion)
  - Volatility Breakout / Volatility Targeting Overlay
  - Simple "ML wrapper" around a baseline signal (great for AI angle)

Prompt 2 Results - Gemini 3

- Foundational quantitative strategies
  - Trend-Following (Momentum)
  - Mean Reversion (Statistical Arbitrage)
  - Pairs Trading (Relative Value)
  - Sentiment-Driven Macro

Prompt 3 Results - ChatGPT 5.2

- Beginner-friendly NLP/sentiment trading strategies
  - News sentiment → trade an index (daily/weekly)
  - Central bank statements (FOMC) sentiment → trade rates / equity proxies
  - SEC filings / earnings text sentiment → trade cross-section (weekly/monthly horizon)

Prompt 3 Results - Gemini 3

- Few entry-level strategies
  - Central Bank Sentiment (Systematic Macro)
  - Commodity Supply/Demand News
  - 10-K/10-Q "Risk Factor" Drift (Quant Equities)
  - The "Analyst Tone" Leading Indicator (Short-Term)

ChatGPT vs Gemini

- Results from all three prompts show overlapping strategies.
- For all three results, not all strategies match.
- For detailed results, refer docs, chatgpt and gemini logs.

Background

- Commodity markets (especially energy) are strongly influenced by exogenous news
- A large portion of relevant information is unstructured text
- Recent advances in financial domain–specific language models (e.g., FinBERT-style architectures) enable systematic extraction of sentiment signals from financial texts
- Sentiment signals can be integrated as an additional information channel alongside price, volatility, and term-structure features

Asset Choice

- Focus on energy-related futures (crude oil for now)
- Advantage over stocks:
  - Slower information decay
  - Stronger economic linkage
  - More coherent market-wide sentiments
  - Lower cross-sectional noise
  - Daily frequency aligns well for our purpose

- WTI Crude Oil Futures (NYMEX) and Brent Crude Oil Futures (ICE)

Sentiment Analysis

- We use FinBERT to implement sentiment analysis
- FinBERT:
  - Financial Sentiment Analysis with BERT
  - a pre-trained NLP model to analyze sentiment of financial text

Current Implementation

- Uses **FinBERT** for sentiment analysis
- Processes **oil-related futures news**
- News collected via **Google News RSS**
- Sentiment computed at the **headline level only**
- Aggregates sentiment to produce a **daily market sentiment summary**


```
Article 1: Oil Futures Fall on Easing Tensions But Risks Remain - The Wall Street Journal
Link: https://news.google.com/rss/articles/CBMisgNBVV95cUxQd25YSXp2VVZqMTRkZkVtU3FiLW9IeFZ5TFNrY210NmlLMVNBdENLVH
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Published: Mon, 02 Feb 2026 19:58:00 GMT
Sentiment: Neutral (Confidence: 1.00)

Article 2: Metals, crude oil dive in broad commodities market tumble - Reuters
Link: https://news.google.com/rss/articles/CBMilgFBVV95cUxNbnZxMnYwUE5sWFBjMktQdEh6ZDFKSmEzbHFqYl85ZkRGQUZmWXQwc0
FHN1lXaE51OVNyRnlWa3QxSXdZUXA1VldBeG8yQTBJd1RaNzZVZnBJMXEyd0ZIZm0zOHo2RVo0Tlprem1vWjRLQ055bEx5Y1pCbjNmMXZSellXMTU
4TVJwNVNRUUN0aXBzU0YyUVE?oc=5
Published: Mon, 02 Feb 2026 11:44:16 GMT
Sentiment: Neutral (Confidence: 1.00)

Article 3: Top energy expert puts odds of U.S. attack on Iran at 75% - 'this one is real' - Fortune
Link: https://news.google.com/rss/articles/CBMirgFBVV95cUxObm52ZUxUOEJjeVNSNTdWZENYQjBELTNRVVZ2cWZFc3g4dHI5Wm40Mk
5DVF9BbzZoNkVuXzE4QVpXOGhaRHg0dlVRNlJMbEJUX0tHVjN1WkdBOGl6MjExd2xjZlJLcDE3Q3lwV2xHTDJ6ZW1faGthcU9HdnloMjJFWERMX2N
JWnVpZEkyTldXUlluQ3hkaWtVUUx6X2Q4OFh2cmQtMG5pbThFUlZYTkE?oc=5
Published: Sun, 01 Feb 2026 21:53:00 GMT
Sentiment: Positive (Confidence: 1.00)

Article 4: Oil settles down over 4% on US-Iran de-escalation, easing supply worries - Reuters
Link: https://news.google.com/rss/articles/CBMioAFBVV95cUxQS2d3NklrTjVqYjhaZHBoTmZicGF2QWNtWDg4NnJWQkZ0Xy1QcUxCa0
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FaG8weWdVRzhOMUhaTHJUektCblJENXpFNEhY?oc=5
Published: Sun, 01 Feb 2026 23:20:33 GMT
Sentiment: Neutral (Confidence: 1.00)

Article 5: Oil Futures Fall on Easing Tensions But Risks Remain - The Wall Street Journal
Link: https://news.google.com/rss/articles/CBMisgNBVV95cUxOWHFJbkgtRDM2Y3FwdDBjUHlqZ2tGTjBjZjVTVzNqVWt0X2hfdGwwRE
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VMV1R4NlQyNWtUbkFWZlZMakNLLTRSenBoMXBrVVlfaHE2Q1l0aXRZUktrUHFodw?oc=5
Published: Mon, 02 Feb 2026 19:58:00 GMT
Sentiment: Neutral (Confidence: 1.00)

Article 6: U.S.-India Trade Deal Puts Oil and Russian Crude at the Core - Crude Oil Prices Today | OilPrice.com
Link: https://news.google.com/rss/articles/CBMitAFBVV95cUxPRlhBUDZBM2k0OEU3RWxuLUNQVUJndld2TF9vdnNkbVFPVkcwTmNxMF
RjRVNEOWlGSnd2ekY0WnZXQlpLM1hBbkJ0ak9BZzRTMHROdmk1OHp5R0htTHdIZGo1ZzZGbDhFMkJrREsxaWJhNE9LeDhwNFVZc3dYVmw4WkJacjZ
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Published: Mon, 02 Feb 2026 19:45:00 GMT
Sentiment: Positive (Confidence: 1.00)
```

Sentiment Analysis(Examples)

- Crude Oil

Article : Too energy expert puts odds of U.S. attack on Iran at 75% - 'this one is real' - Fortune

Link: https ://news .00oale.com/rss/articles /CBMiraFBWV95CUXObm52ZUxUOE]ieWNSNTdWZENYOiBEL TNRWZ2CWZFC304dHI5Wn40Ml5DVF9BbzZONkVuX2E40VpXOGhaRHgOd VRNLJMbEJUXOtHV NIMKdBOGL6MEXd2X]ZLJLCDE303 LW2XHTDJ6ZMI faGthCU9Hdn LOM]JFVERMX2NJWnVpZEkyTldXU1luQ3hkawtVUUx6X2Q40Fh2cmQtMG5pbThFU1ZYT kE?oc=5

Published:Sun,01 Feb 2026 21:53:00 GMT

Sentiment:Positive (Confidence:1.00)

Sentiment Analysis(Examples)

- Gold

Article : Gold and silver prices fall but FTSE 100 hits record hiah - BBC

Link:https://news.900gle.com/rss/articles/CBMiWkFVX3LxTE82V3J6YO5fSjE00U9EOHd0ZnFabEp1eXZWWGZUbnNSazJ0ZWpBROVHalxYcWcOYldLbFZ0SkM3LXNZb1F1aVdpaXhXNUVoekRjZkgwZVA0WXdMUQ?oc=5

Published:Mon,02 Feb 2026 19:28:14 GMT

Sentiment:Negative (Confidence:0.68)

Investment Thesis

News sentiment reflects market expectations about future supply, demand, and risk premia in oil and energy markets.

Trading styles

- Event-driven vs. **continuous rolling**
- **Directional** vs. relative
- **Independent assets** vs. cross-sectional ranking

For an simple beginning, we chose continuous rolling, directional, and independent assets trading style.

Strategy Design

- After we get the daily sentiment scores, we:
  1. Calculate moving average of sentiment scores with different horizons:
     - $s_t^{(1)} = \text{oil sentiment mean over last 1 day}$
     - $s_t^{(3)} = \text{oil sentiment mean over last 3 days}$
     - $s_t^{(7)} = \text{oil sentiment mean over last 7 days}$
     - $c_t^{(7)} = \text{number of oil news articles in last 7 days}$
  2. Get the daily market data: settlement price, return, and volatility
  3. Define a raw daily sentiment signal as:

$$S_t = 0.2 \cdot s_t^{(1)} + 0.3 \cdot s_t^{(3)} + 0.5 \cdot s_t^{(7)}$$

Strategy Design (cont.)

4. Define a news intensity filter so that days with small amount of news get lower weights:

$$I_t = \log(1 + c_t^{(7)})$$

$$\tilde{S}_t = S_t \cdot I_t$$

5. Compute a rolling z-score:

$$Z_t = \frac{\tilde{S}_t - \mu_t}{\sigma_t^{(S)}}$$

where μ_t, $\sigma_t^{(S)}$ are rolling mean and std of $\tilde{S}$ over last 120 days.

Strategy Design (cont.)

6. Define the following threshold trading rules:

- If current position == 0 (neutral):
  - If $Z_t > 0.5$: go long
  - If $Z_t < -0.5$: go short
- If current position == +1 (long):
  - If $Z_t < 0.1$: exit to neutral
- If current position == -1 (short):
  - If $Z_t > -0.1$: exit to neutral

Strategy Design (cont.)

7. Generate desired size of position with volatility targeting:

- Let $d_t \in \{-1, 0, +1\}$ be the position signal for day t.
- Set a target daily portfolio volatility, target_vol, and a maximum position size w_max
- Calculate the position scale:

$$w_t = d_t \cdot \min\left(\frac{\text{target_vol}}{\sigma_t}, w_{\max}\right)$$

8. Include transaction costs:

- Define: $$\text{turnover}_t = |w_t - w_{t-1}|$$
- Assuming transaction cost per unit of position is 2 basis points, the daily cost is

$$\text{cost}_t = \text{turnover}_t \cdot 0.0002$$

Strategy Design (cont.)

9. Calculate the net return:

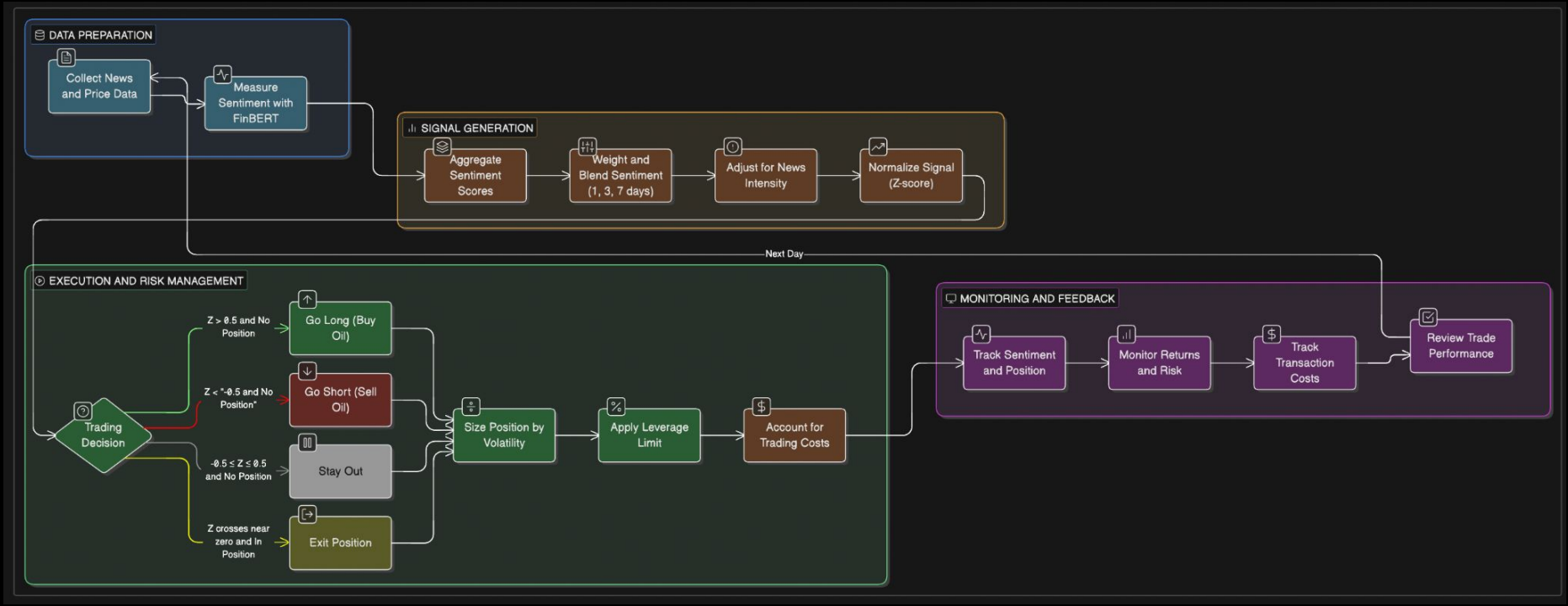
- Suppose daily return of the asset is r_t, then the strategy's daily return is

$$\text{strategy_return}_t = w_{t-1} \cdot r_t - \text{cost}_{t-1}$$

Project Statement

The goal of this project is to build and evaluate a **sentiment-based** trading pipeline for oil futures, integrating financial news with market price data to test whether news sentiment provides predictive value for trading decisions.

Project Workflow



Key Steps

1

Data Collection & Preparation

Financial news and historical price data are collected and aligned at a daily frequency.

This step ensures that unstructured text (news) and structured data (prices) are temporally consistent and ready for modeling.

2

Sentiment Modeling

A finance-specific language model is applied to news articles to extract sentiment at the article level, which is then aggregated to daily sentiment measures and stored for reproducibility.

3

Sentiment Signal Construction

Daily sentiment is transformed into trading signals using rolling averages, news-intensity adjustment, and normalization to ensure stability and comparability over time.

4

Trading Strategy Design

Daily sentiment is transformed into trading signals using rolling averages, news-intensity adjustment, and normalization to ensure stability and comparability over time.

5

Backtesting & Evaluation

The strategy is evaluated using historical price data, accounting for transaction costs and risk, and compared against baseline strategies.

Data (Oil)

News data :

Reuters – Energy / Oil :
https://www.reuters.com/business/energy/

OilPrice.com :
https://oilprice.com/

Price data :

Yahoo Finance – WTI Crude Oil Futures (CL=F)
https://finance.yahoo.com/quote/CL%3DF/

Investing.com – Crude Oil Futures
https://www.investing.com/commodities/crude-oil

Deliverables and Task division

Integrating More News & Price Data

We are integrating additional **news data and price data** into the **existing daily news-to-trading flow**.

Why this is needed:

More news data → stable daily sentiment, rolling windows, robustness
Price data → returns, volatility, backtesting, and risk control.

All data are aligned at a **daily frequency**

Harris

Storage

Storage is introduced to:

Ensure reproducibility,Support rolling-window and Enable debugging and analysis

Storage focuses on **sentiment outputs and metadata**, not raw text.

Medha

Handling Data Duplicity

Financial news is often syndicated or repeated.

Why this matters:

Prevents duplicate articles from overweighting sentiment
Ensures sentiment reflects **unique information**, not volume.

Deduplication is applied **before sentiment aggregation** and combined with news-intensity adjustment.

Aswath

Deliverables and Task division

Exploring Sentiment Models

Model choice is still under exploration.

All models are evaluated using the **same data, signals, and strategy**.

Yue

Full-Document Sentiment

We are exploring:

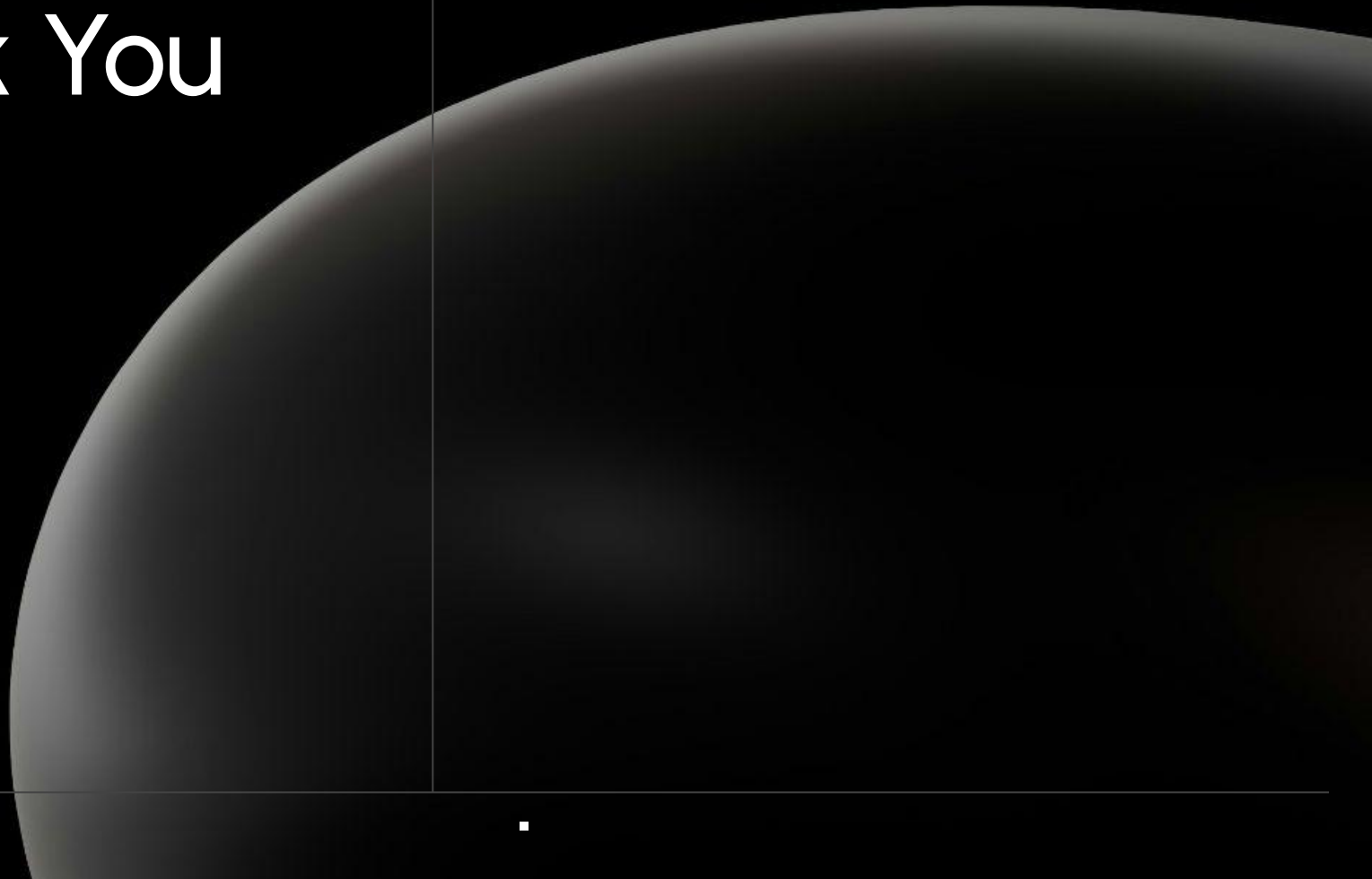
Full-article vs headline-only sentiment, Truncation vs chunking
Compute and storage tradeoffs

This is an **extension**, not a replacement, for the headline-based baseline.

Medha

Aswath

LLMs Used

- ChatGPT 5.2
- Gemini 3
- Claude Sonnet 4.5
- FinBERT (/home/ypan81/models/finbert-tone)

Thank You